

Notes

18.06 Problem Set ?

1.0 / 100

Diff. Problem

12. $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

is A & A^T do

not generally have the

same nullspace as the

free variables may change

$x_1 + x_3 = 0$

$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$x_3 = 0$

$x_1 + x_2 = 0$ are independent

$x_2 + x_3 = 0 \Rightarrow$ no solution

$A = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 9 \end{bmatrix}$

more free variables

$\Rightarrow$ more solutions

$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 3 & 5 \\ 0 & 0 & 1 & 9 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 3 & 1 \\ 4 & 5 & 9 \end{bmatrix}$

Free x_4

Free = N/A

36. $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} x & y & z \\ u & v & w \end{bmatrix}$

$R_1 = \begin{bmatrix} a & b & c \\ 0 & e-bf & f \end{bmatrix}$ $R_2 = \begin{bmatrix} d & e & f \\ 0 & v-gf & 2 \end{bmatrix}$

$R_1 + R_2 = \begin{bmatrix} a & b & c \\ 0 & e-bf & f \end{bmatrix}$

$A_1 + A_2 = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

The condition $\text{ref}(A_1 + A_2) = R_1 + R_2$

$\Rightarrow A_1 = R_1, A_2 = R_2$ False

$R_1 - R_2 \neq \text{ref}(A_1 - A_2)$ since

row reduction is not addition

False

* All of Section 3.2 is different

Diff.

$\downarrow$

37. $A = \begin{bmatrix} 1 & 5 & 4 \\ 2 & 10 & 8 \\ 3 & 15 & 12 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 10 & 15 \\ 4 & 8 & 12 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 5 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Rank = 1

$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Rank = 1

$\Rightarrow \begin{bmatrix} 1 & 5 \\ 0 & 10 \end{bmatrix}$ will always lead

to $\text{rank}(A) = \text{rank}(A^T)$. For non-invertible, the dimension always incorporates the change.

Diff.

$\downarrow$

$m \times n$

38. a. $r \in \mathbb{R}^m, b \in \mathbb{R}^m, r \in \mathbb{R}^n$

$\Rightarrow c \in \mathbb{R}^m, r \in \mathbb{R}^n$

Diff.

38. $(U \vec{0}) = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $(U \vec{2}) = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} R_1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} R_1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

Diff.

39. $\text{rank}(A)$

$\begin{bmatrix} 1 & 1 & 5 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 5 \\ 0 & 1 & 4 \end{bmatrix}$

Rank = 2

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Rank = 2

40. A matrix when A is

a square matrix with

$\det(A) \neq 0$

All of section 3.3 is also different

THEOREM

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 $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

a. $C(A) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$ b. $C(U) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \Rightarrow$ Dimension of $\begin{bmatrix} 2 \end{bmatrix}$ $\Rightarrow$ Dimension of $\begin{bmatrix} 2 \end{bmatrix}$

c. $C(A^3) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ d. $C(U^3) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Dimension of $\begin{bmatrix} 1 \end{bmatrix}$ Dimension of $\begin{bmatrix} 2 \end{bmatrix}$
 $C(A)$ and $C(U)$ share the same space

P.H.
 $\downarrow$

35. $P(x) \in \mathbb{R}_3$ $P(x) = 0$
 $\Rightarrow ax^3 + bx^2 + (c+d)x = 0$
 $\Rightarrow ax^3 + bx^2 + cx + dx = 0$
 for $\{(a,b,c,d) \in \mathbb{R}^4\}$

* Since all problems within the post are different, I resorted to googling it separately. Therefore a more fitting grade is N/A; but this doesn't give the same experience.

36. $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

f. numbers $c, d \in \mathbb{R}$ s.t. $(c, d) \in C(A)$

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

A basis would be

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

e) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

P.H.
 $\downarrow$

36. $(a,b,c,d) := a+c+d=0$
 $\Rightarrow c+c+(-2c)$
 $\therefore$ for all permutations of $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
 $(a,b,c,d) := b+c+0$
 $\Rightarrow d-d$
 $= \begin{bmatrix} d \\ -d \end{bmatrix}$ or $\begin{bmatrix} -d \\ d \end{bmatrix}$
 $(a,b,c,d) := c+2d$
 $\Rightarrow \frac{c}{d} = 2$

$\therefore$ The basis of $\begin{bmatrix} e \\ e \\ e \\ 2e \end{bmatrix}$ for $e \in \mathbb{R}$

Dimension for SAT is $\begin{bmatrix} 2 \end{bmatrix}$ as there are 2 common elements